



Wings for Science

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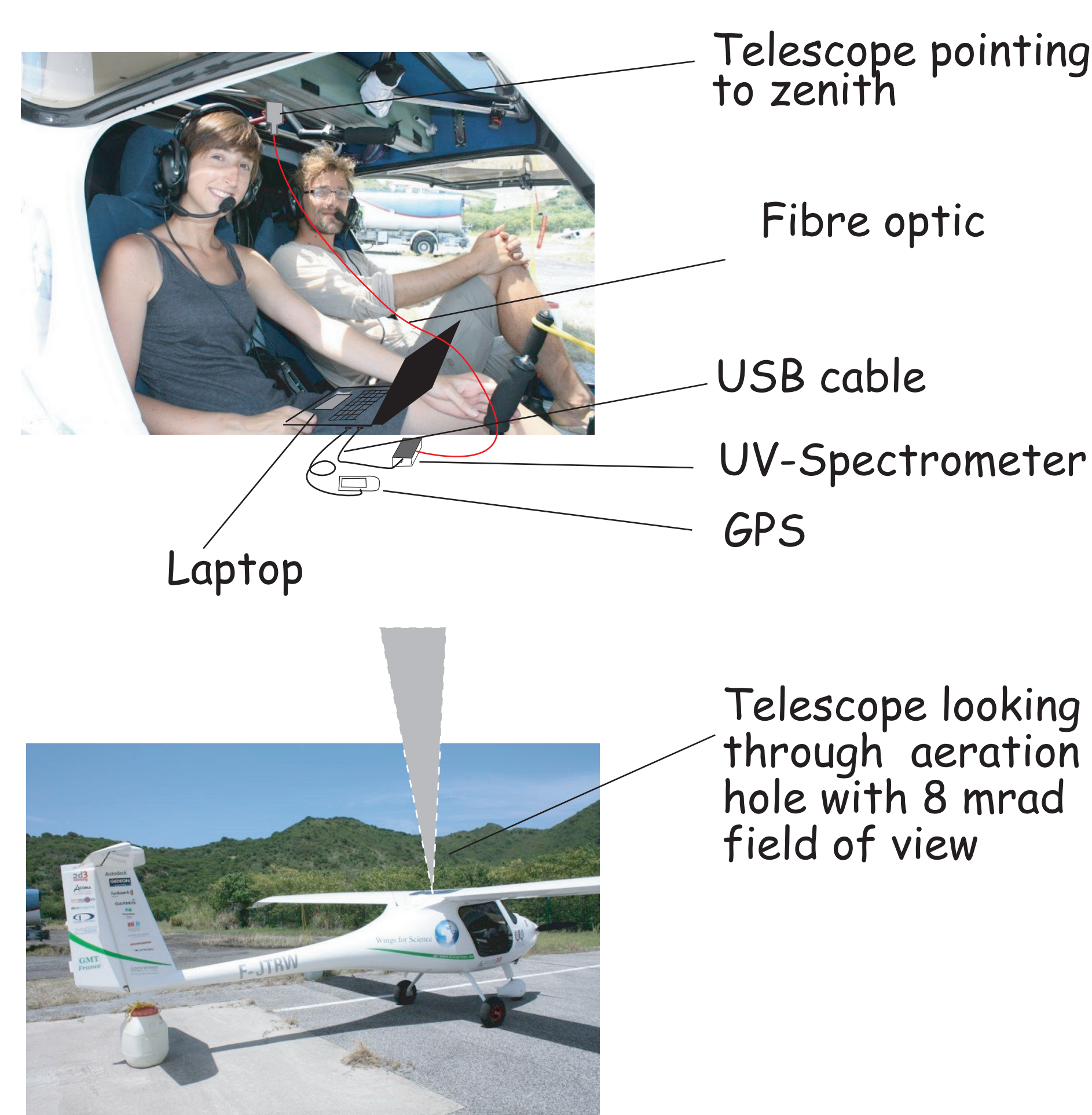
Brief History of Krakatau Volcanic Activity

Situated between Java and Sumatra islands, this volcano is among the two most deadliest volcanoes known on earth, both in Indonesia, with 36600 fatalities in 1883. The tragic record being that from Tambora in 1815 killing more than 60000 people. Krakatau eruption was classified VEI 6, equivalent to 200 megatons of TNT. It was heard up to 3,500 km from source. The eruption tore apart the Krakatau island-volcano, generation a huge tsunami across the region, induced spectacular sunset throughout the world for many months and caused a global temperature fell by 1.2 °C. In the past 130 years, more than 40 eruptive activities are reported to occur on Krakatau, leading to the formation of Anak Krakatau (Child of Krakatau). In last decade, with continuous monitoring means and satellite observations, around 15 eruptive events were detected, indicating that Krakatau is one of the most active volcanoes in Indonesia. These eruptive manifestation ranged from output of ashy plumes to lava flows, originated from flanks and extended to the coast. Will the new and very active Anak Krakatau cone lead to another 1883? The question remains open.

We present here for the first time, the estimates of SO₂ emission rate from Anak Krakatau using DOAS (Differential Optical Absorption Spectroscopy), carried onboard the **Wings for Science light aircraft**.

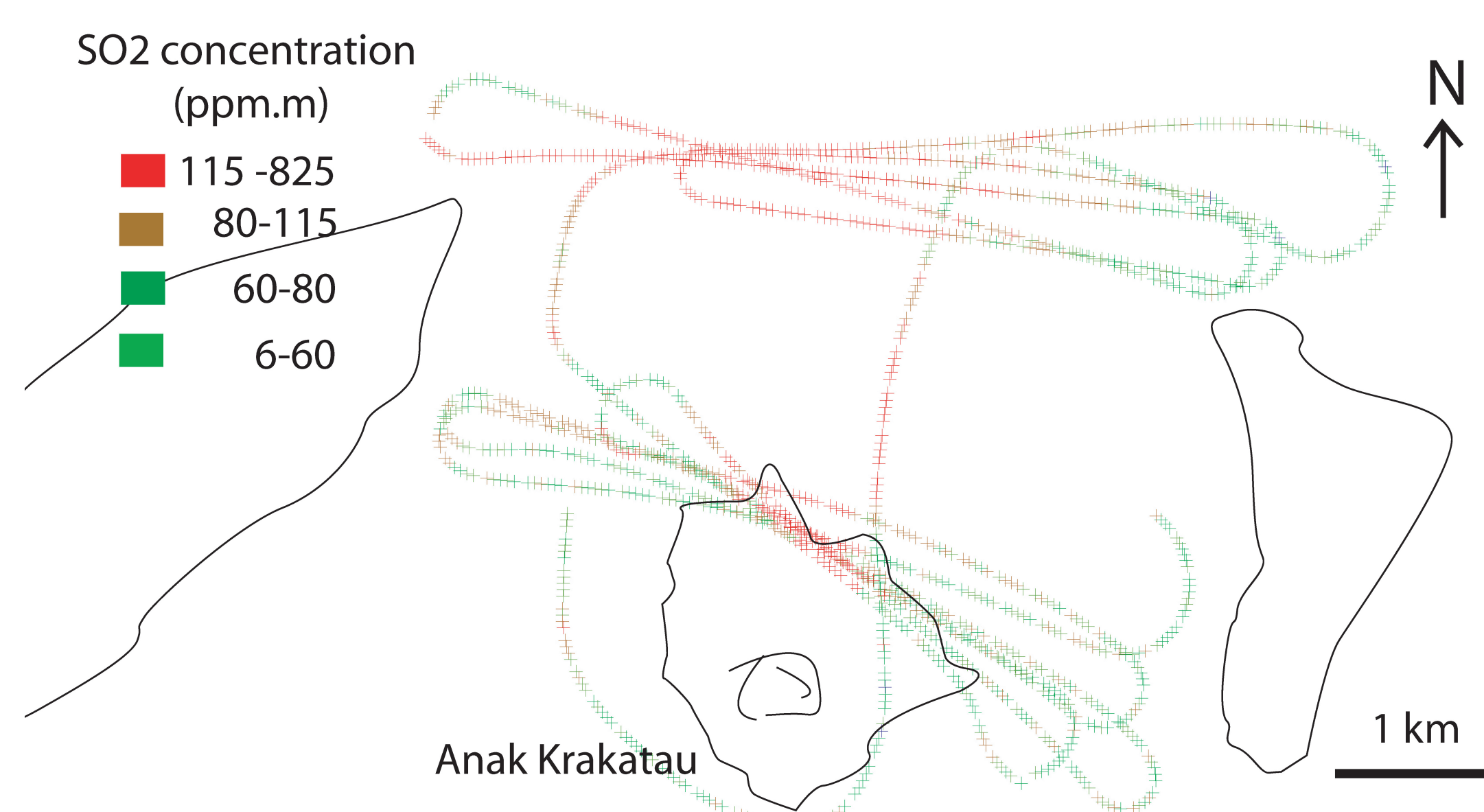
Methods

The measurements were carried out on April 2, 2013. The instrument used was a USB4000 UV spectrometer (Serial # 2046) from Ocean Optics with a spectral range of 290-440 nm and a spectral resolution of 0.3 nm. The device was coupled to a vertically pointing telescope of 8 mrad field of view with an optic fibre bundle. A USB cable connected the spectrometer to a laptop computer, providing power and data transfer. Software control of the USB2000 was achieved using Jscript executed in DOASIS software. The exposure time for individual spectra 200 ms and we co-add 4 spectra to enhance the signal-to-noise ratio. The position of each spectrum was determined from the log of continuous recording GPS unit.



SO₂ column amount was retrieved according to standard differential optical absorption spectroscopy procedure following Beer-Lambert law that describes the absorption of electromagnetic radiation by matter. Acquired spectra were first corrected for blind and offset using data from blind UV interval (below 295 nm). Resulting spectra were then divided by reference spectrum (obtained outside volcanic plume). Finally, a nonlinear, least squares fitting procedure were applied, using SO₂ absorption spectrum of known amount, obtained by convolution of high spectral resolution laboratory spectrum with the response function of the spectrometer. SO₂ fluxes were calculated by multiplying each retrieved concentration by the distance traversed perpendicularly to the plume transport direction during the acquisition of that data point. These products were then sum across the entire plume and multiplying the total by estimated plume speed (assuming equivalent to wind speed in our case). Wind speed was obtained from aircraft navigation system.

Results



10 traverses under the plume of Anak Krakatau were carried out during this work at distance ranging from 600 m to 5 km from source. The aircraft evolved at about 200 m above sea level.

The figure to left shows the flying route and the SO₂ concentration encountered. High SO₂ concentrations appear aligned in the Nord-South direction on the traverses, indicating the plume dispersion to the north. The plume width change from 1.1-1.2 km at 0.6-1 km from source to 3-4 km width 3-5 km away, whilst the mean SO₂ concentration per spectrum decreased about 3x at 5 km away from the source.

The table below summarize the SO₂ flux results obtained in this work. SO₂ emission rate fluctuates between 1.0 to 3.8 kg/s with a mean emission rate of 2.1 kg/s.

Traverse number	Start time LT (LT=UT+6)	Dist. From source (km)	Plume width (km)	Mean SO ₂ column amount (mg/m ³)	SO ₂ flux	
					(kg/s)	t/d
Trav. 1	10:15:37	3.2	4.6	177.3	3.8	329.0
Trav. 2	10:17:55	5.2	3.6	210.8	3.3	282.3
Trav. 3	10:20:02	4.9	4.1	145.0	3.0	256.5
Trav. 4	10:22:44	0.6	0.9	252.6	1.0	88.5
Trav. 5	10:24:15	0.7	2.0	232.2	1.9	169.6
Trav. 6	10:25:36	0.7	1.7	290.0	2.3	198.3
Trav. 7	10:28:08	0.8	1.1	230.7	1.2	100.3
Trav. 8	10:30:11	0.7	1.1	295.9	1.3	117.1
Trav. 9	10:31:54	1.0	2.2	244.5	2.3	201.5
Trav. 10	10:34:10	0.8	1.2	228.2	1.2	101.9
Mean SO ₂ emission rate = 2.1 kg/s (184.5 t/d)						

Discussion and conclusion

DOAS airborne measurements results indicate that Anak Krakatau is a significant source of magmatic output, releasing more than 180 tons of sulfur dioxide daily into to the atmosphere. This figure is higher than the 140 t/d of SO₂ release from the well known Merapi volcano, considered as the biggest source of SO₂ release in Indonesia outside eruptive manifestation. Anak Krakatau whose SO₂ emission was never taken into account in global volcanic SO₂ emission budget deserves largely its integration as it may be positioned in the middle of the 49 listed volcanoes considered as the biggest sources of SO₂ on earth (Andres An Kasgnoc, 1998). Assuming that the figure obtained in this work is representative of the long term emission, then Anak Krakatau can release more than 67 kt of sulfur dioxide annually into the atmosphere, that is 0.5 % of the global annual volcanic SO₂ budget.